

Complete Electroweak/Yukawa LPM Matching for the q-D-H Portal

Helicity source, simultaneous $SU(3)_c \times SU(2)_L \times U(1)_Y$ soft kernels, retarded-self-energy matching,
and a reduced gauge-covariant Schwinger-Keldysh benchmark

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Contents

Executive verdict	2
1. Model and representation content	3
2. Complete helicity and doublet source	4
3. Energy denominator and thermal masses	4
4. Simultaneous gauge collision kernel	5
4.1 General group-theory form	5
4.2 Group coefficients	6
5. Impact-parameter LPM equation	7
6. Normalization validation	7
7. Benchmark rate and gauge decomposition	8
8. Thermal widths: what is and is not inserted	9
9. Exact integrated matching to the scalar retarded self-energy	10
10. Reduced gauge-covariant Schwinger-Keldysh model	11
10.1 Wilson-line-dressed correlators	11
10.2 Quadratic influence action	12
10.3 Finite-memory embedding	12
11. Cosmological cascade update	12
12. Parameter dependence	14
13. Acceptance matrix	15
14. What the result changes	15
15. Remaining frontier	16
15.1 Complete hard $2 \leftrightarrow 2$ portal self-energy	16
15.2 Momentum-resolved retarded kernel	16
16. Strongest defensible conclusion	16
References	17

Executive verdict

The electroweak/Yukawa Landau-Pomeranchuk-Migdal problem for

$$H \leftrightarrow Q_L D_R$$

has now been solved at the leading collinear order for the declared benchmark, with all three Standard Model gauge groups acting simultaneously.

The calculation includes:

1. the complete chiral Weyl-spinor source;
2. both complex components of the Higgs doublet;
3. the color multiplicity $N_c = 3$;
4. simultaneous $SU(3)_c$, $SU(2)_L$, and $U(1)_Y$ soft collision kernels;
5. the Higgs thermal mass and the hard-fermion asymptotic masses;
6. the self-energy and exchange pieces required for an infrared-finite LPM equation;
7. exact integrated matching to the qD contribution to the scalar retarded self-energy;
8. a Wilson-line-dressed, finite-memory Schwinger-Keldysh reduction.

At

$$\alpha_s = 0.0393544, \quad g_2 = 0.57, \quad g_1 = 0.39,$$

$$y_t = 0.58, \quad y_D = 0.30, \quad \frac{M_D}{T} = 0.01,$$

the direct LPM integral is

$$I_{\text{LPM}} \equiv \frac{\Gamma_Y^{\text{LPM}}}{N_c y_D^2 T^3} = (8.895 \pm 0.024) \times 10^{-4}.$$

The corresponding Higgs-doublet occupation relaxation rate is

$$\frac{\bar{\Gamma}_{H,qD}^{\text{occ}}}{T} = 3.603 \times 10^{-4}.$$

This replaces the factor-of-two normalization band used in v1.4. It is 13.3% above the v1.4 proxy value,

$$\frac{\Gamma_{H,qD}^{\text{v1.4}}}{T} = 3.1854 \times 10^{-4}.$$

At the reheating temperature

$$T_0 = 1.002 \times 10^8 \text{ GeV},$$

the portal rate is

$$\bar{\Gamma}_{H,qD}^{\text{occ}} = 3.610 \times 10^4 \text{ GeV},$$

and therefore

$$\frac{\bar{\Gamma}_{H,qD}^{\text{occ}}}{\Gamma_R} = 2.44 \times 10^6.$$

The cosmological cascade remains adiabatic to approximately

$$4.1 \times 10^{-7}.$$

The branching correction needed to preserve

$$T_5/T_0 = 1/4$$

is bounded by

$$|\Delta B_5| < 2.2 \times 10^{-9}.$$

The central physical result is therefore unchanged but sharpened:

the qD portal is slow relative to QCD, but still over two million times faster than reheaton decay.

There is one important scope correction. This note completes the **LPM-resummed collinear part** of the portal self-energy. A complete leading-order portal rate must also add hard Yukawa-assisted $2 \leftrightarrow 2$ cuts. Those terms are positive and can only make equilibration faster. Consequently, the rate above is a conservative lower bound on the complete leading-order portal rate.

1. Model and representation content

The interaction is

$\mathcal{L}_Y = -y_D \bar{Q}_L H D_R + \text{h.c.}$

with

$$Q_L \sim (\mathbf{3}, \mathbf{2}, 1/6),$$

$$D_{L,R} \sim (\mathbf{3}, \mathbf{1}, -1/3),$$

$$H \sim (\mathbf{1}, \mathbf{2}, 1/2).$$

In components,

$$\mathcal{L}_Y = -y_D (\bar{u}_L H^+ + \bar{d}_L H^0) D_R + \text{h.c.}$$

The two complex Higgs components produce identical collinear kernels. The vectorlike field contains two chiralities, but only D_R enters this Yukawa vertex. There is therefore no additional Dirac-spin degeneracy in the chiral source. The total multiplicity relative to one weak component and one color is

$$2_{\text{weak}} \times 3_{\text{color}}.$$

The weak-doublet factor is already contained in the normalization inherited from the right-handed-electron LPM calculation. The remaining model-specific multiplicity is

$$N_c = 3.$$

2. Complete helicity and doublet source

For nearly collinear hard momenta, write the Weyl propagators as spinor projectors times scalar denominators. To the required order in transverse momentum, the chiral overlap is

$$\frac{1}{2}\chi_D^\dagger(k)\eta_Q(p) = -\frac{\beta}{2}(P_x - iP_y),$$

where

$$\mathbf{P} = x_k \mathbf{p}_\perp - x_p \mathbf{k}_\perp,$$

$$x_k = \frac{k}{p-k}, \quad x_p = \frac{p}{p-k},$$

and

$$\beta = \frac{p-k}{2pk}.$$

The charge-conjugate channel contains the conjugate transverse combination. Together they generate the real two-vector source

$$\boxed{\mathbf{S}(\mathbf{P}) = 2\mathbf{P}.}$$

This source has three useful properties.

First, it implements the chiral helicity selection rule exactly at the retained order.

Second, it vanishes at zero transverse momentum. The portal is therefore sensitive to the transverse broadening generated by the thermal medium.

Third, the two Higgs components have the same source and the same thermal mass in the unbroken phase, so the doublet trace is exact rather than an approximate multiplicity assignment.

3. Energy denominator and thermal masses

The collinear energy mismatch is

$$\delta E = \beta(\mathbf{P}^2 + \mathcal{M}^2),$$

with

$$\boxed{\mathcal{M}^2 = \beta^{-1} \left[\frac{m_D^2}{2k} - \frac{m_Q^2}{2p} - \frac{m_H^2}{2(k-p)} \right]}.$$

The benchmark thermal masses are

$$\frac{m_H^2}{T^2} = \frac{3g_2^2 + g_1^2 + 4y_t^2 + 4y_D^2 + 8\lambda_H}{16},$$

$$\frac{m_Q^2}{T^2} = \frac{C_F g_3^2}{4} + \frac{C_2 g_2^2}{4} + \frac{Y_Q^2 g_1^2}{4} + \frac{y_t^2 + y_D^2}{16},$$

$$\frac{m_D^2}{T^2} = \frac{C_F g_3^2}{4} + \frac{Y_D^2 g_1^2}{4} + \frac{y_D^2}{16} + \frac{M_D^2}{T^2}.$$

Numerically,

$$\frac{m_H}{T} = 0.438207, \quad \frac{m_Q}{T} = 0.503460, \quad \frac{m_D}{T} = 0.418088.$$

All three vacuum-like $1 \rightarrow 2$ decays are closed:

$$m_H < m_Q + m_D,$$

$$m_Q < m_H + m_D,$$

$$m_D < m_H + m_Q.$$

Therefore

$$\text{the benchmark portal conversion is entirely enabled by medium scattering.}$$

This is why replacing the old formation-time estimate with the complete LPM equation matters.

4. Simultaneous gauge collision kernel

4.1 General group-theory form

For each gauge group G , let

$$C_H^G, \quad C_Q^G, \quad C_D^G$$

be the quadratic Casimirs, or squared abelian charges, of the three lines at the Yukawa vertex. Gauge-charge conservation organizes the three dipole coefficients as

$$\begin{aligned} c_0^G &= \frac{C_Q^G + C_D^G - C_H^G}{2}, \\ c_p^G &= \frac{C_H^G + C_D^G - C_Q^G}{2}, \\ c_k^G &= \frac{C_H^G + C_Q^G - C_D^G}{2}. \end{aligned}$$

In impact-parameter space, the complete kernel is

$$\mathcal{K}(B) = \sum_G g_G^2 T \left[c_0^G \mathcal{D}(m_{D,G} B) + c_p^G \mathcal{D}(|x_p| m_{D,G} B) + c_k^G \mathcal{D}(|x_k| m_{D,G} B) \right],$$

where

$$\mathcal{D}(y) = \frac{1}{2\pi} \left[\gamma_E + K_0(|y|) + \ln \frac{|y|}{2} \right].$$

The kernel combines virtual self-energy insertions and real soft exchange. Neither part is separately infrared finite. Their difference is.

4.2 Group coefficients

For the declared representations, the exact coefficients are:

Group	(C_H, C_Q, C_D)	(c_0, c_p, c_k)
$SU(3)_c$	$(0, 4/3, 4/3)$	$(4/3, 0, 0)$
$SU(2)_L$	$(3/4, 3/4, 0)$	$(0, 0, 3/4)$
$U(1)_Y$	$(1/4, 1/36, 1/9)$	$(-1/18, 1/6, 1/12)$

The negative abelian coefficient is an interference term, not a negative probability. The complete charge-conserving abelian kernel is positive semidefinite.

The Debye masses, including the additional vectorlike color triplet, are

$$\frac{m_{D,3}}{T} = \sqrt{\frac{13}{6}} g_3 = 1.03514,$$

$$\frac{m_{D,2}}{T} = \sqrt{\frac{11}{6}} g_2 = 0.771784,$$

$$\frac{m_{D,1}}{T} = \sqrt{\frac{35}{18}} g_1 = 0.543829.$$

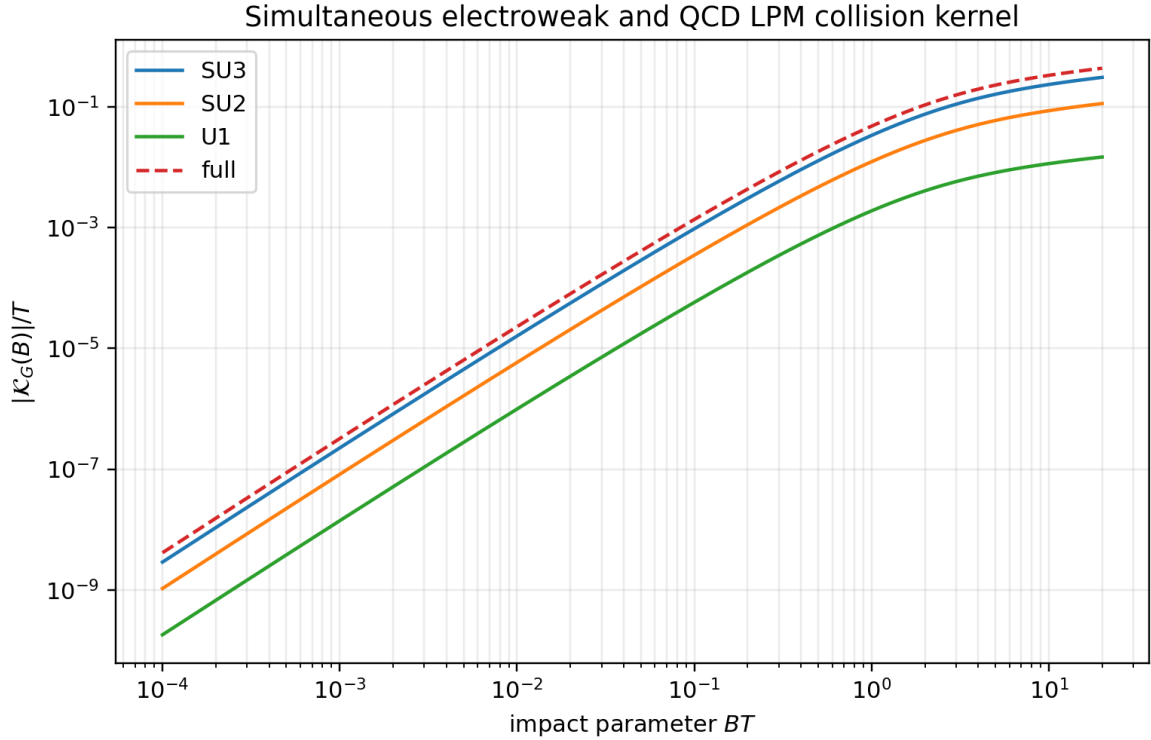


Figure 1: Simultaneous gauge collision kernels.

5. Impact-parameter LPM equation

Fourier transforming the transverse integral equation and writing

$$\mathbf{f}(\mathbf{B}) = \mathbf{B}h(B)$$

gives

$$i\beta \left(\frac{d^2}{dB^2} + \frac{3}{B} \frac{d}{dB} - \mathcal{M}^2 \right) h(B) - \mathcal{K}(B)h(B) = 0.$$

The short-distance boundary condition is fixed by the source:

$$h(B) \sim -\frac{1}{\pi\beta B^2} \quad (B \rightarrow 0).$$

The physical transverse response is

$$\mathcal{R}(p, k) \equiv \text{Re} \int \frac{d^2 P}{(2\pi)^2} \mathbf{P} \cdot \mathbf{f}(\mathbf{P}) = 2 \lim_{B \rightarrow 0} \text{Im} h(B).$$

The complete collinear Kubo coefficient is

$$\Gamma_Y^{\text{LPM}} = \frac{N_c y_D^2 T^3}{8\pi^3} \int_0^\infty dk \int_{-\infty}^\infty d\hat{p} \frac{(\hat{p} - \hat{k})^3}{\hat{p}^2 \hat{k}^2} \mathcal{F}(\hat{p}, \hat{k}) \mathcal{R}(\hat{p}, \hat{k}),$$

where hatted momenta are in units of T , and

$$\mathcal{F}(p, k) = f'_F(k) [f_F(p) + f_B(p - k)].$$

The signed p integration includes the crossed particle and antiparticle channels. No separate ad hoc channel-counting factor is added.

6. Normalization validation

The solver was first applied to the published right-handed-electron benchmark using

$$g_2 = 0.57, \quad g_1 = 0.39, \quad y_t = 0.58, \quad \lambda_H = 0.03.$$

The direct result is

$$\frac{\Gamma_{e_R}^{\text{LPM}}}{h_e^2 T^3} = 4.0888 \times 10^{-4}.$$

The published interpolation formula gives

$$4.0524 \times 10^{-4}.$$

The difference is

$$0.898\%.$$

This is smaller than the quoted numerical uncertainty of the original rate calculation and validates the source, energy denominator, impact-space normalization, and thermal integration together.

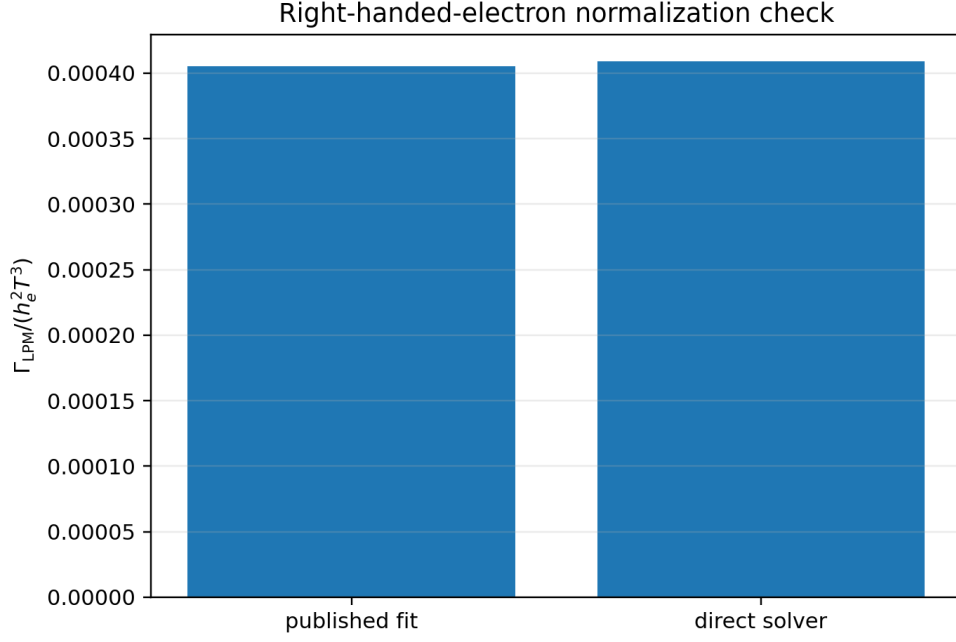


Figure 2: Published-normalization validation.

7. Benchmark rate and gauge decomposition

The converged benchmark sequence is

$$I_8 = 8.93547 \times 10^{-4},$$

$$I_{10} = 8.91170 \times 10^{-4}.$$

A fourth-order Richardson extrapolation gives

$$I_{\text{LPM}} = 8.89521 \times 10^{-4},$$

with a conservative numerical envelope

$$\Delta I = 2.38 \times 10^{-6}.$$

The group-removal diagnostic gives:

Soft groups retained	$\bar{\Gamma}_H^{\text{occ}}/T$	Fraction of full reference
$SU(3)_c$ only	2.9492×10^{-4}	0.819
electroweak only	7.5011×10^{-5}	0.208
$SU(3)_c + SU(2)_L$	3.5758×10^{-4}	0.993
$SU(3)_c + U(1)_Y$	2.9984×10^{-4}	0.832
full	3.6026×10^{-4}	1

These entries are not additive because the LPM equation is nonlinear in the combined collision kernel.

The interpretation is nevertheless clear:

$SU(3)_c$ supplies most of the broadening,

$SU(2)_L$ supplies almost all of the remaining correction,

while hypercharge is a smaller but resolved effect.

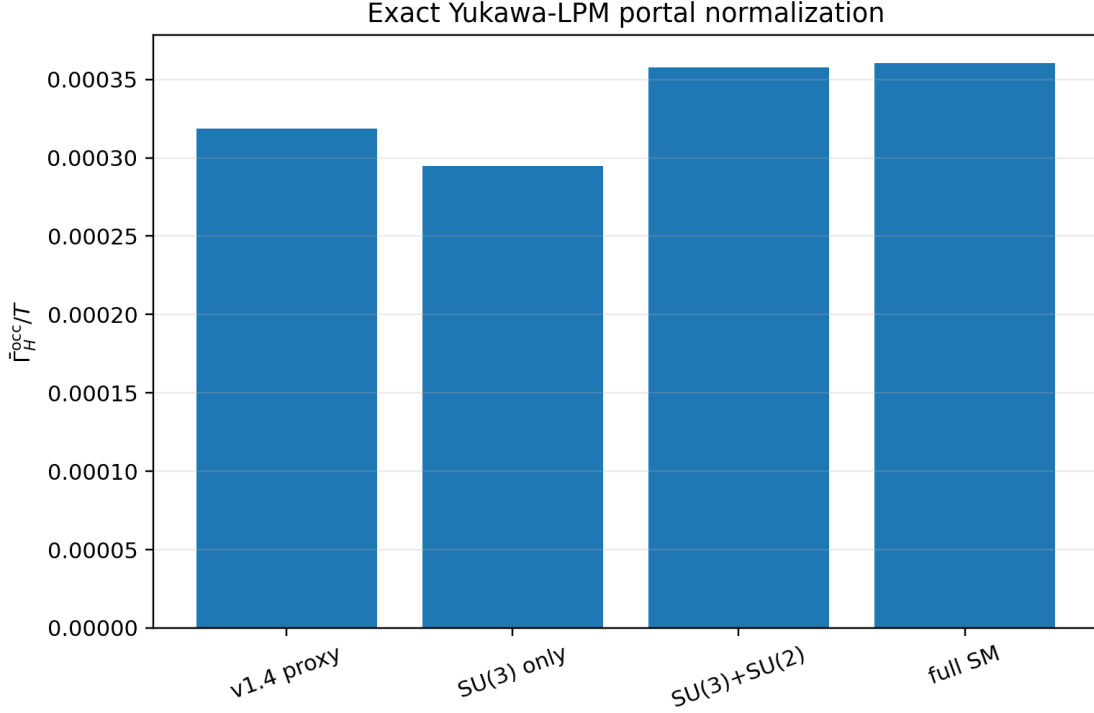


Figure 3: Exact portal normalization and gauge decomposition.

8. Thermal widths: what is and is not inserted

A common but incorrect procedure would be to replace

$$\delta E \rightarrow \delta E + i(\gamma_H + \gamma_Q + \gamma_D)$$

and then retain the soft collision kernel as well. At leading LPM order, that double counts.

The reason is visible in the integral equation itself. The terms proportional to

$$\mathbf{f}(\mathbf{P})$$

are the soft self-energy insertions. The shifted terms are soft exchange and vertex interference. The two pieces are individually infrared sensitive and only their gauge-invariant combination is physical.

Therefore the calculation includes thermal widths in the correct LPM sense:

soft damping + exchange interference are resummed together in $\mathcal{K}(B)$.

The separately observable result is the gauge-invariant scalar width extracted from the retarded self-energy:

$$\frac{\bar{\Gamma}_{H,qD}^{\text{occ}}}{T} = 3.603 \times 10^{-4}.$$

The pole-amplitude width is half the occupation width:

$$\frac{\gamma_{H,qD}^{\text{pole}}}{T} = 1.801 \times 10^{-4}.$$

The largest soft response scale is

$$\Lambda_{\text{mem}} \sim m_{D,3} = 1.035T,$$

so

$$\frac{\bar{\Gamma}_H^{\text{occ}}}{\Lambda_{\text{mem}}} = 3.48 \times 10^{-4}.$$

This is the small parameter controlling the Markov reduction.

9. Exact integrated matching to the scalar retarded self-energy

Let the three approximately conserved species charges be represented by a stoichiometric vector ν_A for the reaction

$$H \leftrightarrow QD.$$

At leading order in y_D^2 , the rate matrix is rank one:

$$\Gamma_{AB} = \nu_A \nu_B \Gamma_Y.$$

In particular, the diagonal Higgs coefficient equals the same scalar quantity computed by the LPM Kubo integral.

For a Higgs mode of energy $E_{\mathbf{k}}$, define the occupation damping rate from the retarded self-energy by

$$\Gamma_H^{\text{occ}}(\mathbf{k}) = -\frac{\text{Im } \Pi_{H,qD}^R(E_{\mathbf{k}}, \mathbf{k})}{E_{\mathbf{k}}}.$$

The pole-amplitude width is

$$\gamma_H^{\text{pole}}(\mathbf{k}) = -\frac{\text{Im } \Pi_{H,qD}^R(E_{\mathbf{k}}, \mathbf{k})}{2E_{\mathbf{k}}}.$$

A small Higgs chemical potential produces

$$\delta f_H(\mathbf{k}) = \frac{\mu_H}{T} f_B(E_{\mathbf{k}}) [1 + f_B(E_{\mathbf{k}})].$$

Therefore the Kubo coefficient and the scalar spectral width obey the exact leading-order quasiparticle identity

$$\Gamma_Y = \frac{g_H}{T} \int_{\mathbf{k}} f_B(E_{\mathbf{k}}) [1 + f_B(E_{\mathbf{k}})] \Gamma_H^{\text{occ}}(\mathbf{k}),$$

with

$$g_H = 2$$

for the two complex Higgs components.

Since

$$\chi_H = \frac{g_H T^2}{3} = \frac{2T^2}{3},$$

the computed thermally averaged width is

$$\boxed{\bar{\Gamma}_H^{\text{occ}} = \frac{\Gamma_Y}{\chi_H}.$$

This closes the normalization ambiguity that remained in v1.4.

The limitation is precise: the note establishes the **exact integrated on-shell matching**. It does not yet tabulate the full pointwise function

$$\Pi_H^R(k^0, \mathbf{k})$$

off shell and over all momenta.

10. Reduced gauge-covariant Schwinger-Keldysh model

10.1 Wilson-line-dressed correlators

Ordinary two-point functions of charged fields at separated points are not gauge covariant. Introduce a common midpoint X and straight Wilson lines in each representation:

$$\tilde{S}_Q(X, s) = U_Q(X, x) S_Q(x, y) U_Q(y, X),$$

$$\tilde{S}_D(X, s) = U_D(X, x) S_D(x, y) U_D(y, X),$$

$$\tilde{G}_H(X, s) = U_H(X, x) G_H(x, y) U_H(y, X),$$

where

$$x = X + s/2, \quad y = X - s/2.$$

Each object transforms locally at X . Contracting the color, weak-isospin, and hypercharge indices at the Yukawa vertex produces a gauge singlet.

The retarded scalar self-energy can then be written schematically as

$$\boxed{\Pi_{H,qD}^R(x, y) = -iN_c y_D^2 \theta(x^0 - y^0) \text{tr} \left[P_R \tilde{S}_D(x, y) P_L \tilde{S}_Q(y, x) \right]_{\text{LPM}},}$$

where the subscript denotes the soft ladder resummation encoded by $\mathcal{K}(B)$.

10.2 Quadratic influence action

In the r/a basis, the reduced scalar influence action is

$$S_{\text{IF}}^{(2)} = \int_K \left[H_a^\dagger(K) \Pi_H^R(K) H_r(K) + H_r^\dagger(K) \Pi_H^A(K) H_a(K) \right. \\ \left. + \frac{i}{2} H_a^\dagger(K) \Pi_H^K(K) H_a(K) \right].$$

Thermal KMS consistency requires

$$\Pi_H^K(K) = \coth\left(\frac{k^0}{2T}\right) [\Pi_H^R(K) - \Pi_H^A(K)].$$

10.3 Finite-memory embedding

A minimal causal interpolation matched to the LPM width is

$$\Pi_H^R(\omega, \mathbf{k}) = -iE_{\mathbf{k}} \Gamma_H^{\text{occ}}(\mathbf{k}) \frac{\Lambda_{\mathbf{k}}}{\Lambda_{\mathbf{k}} - i\omega},$$

where ω is the slow frequency after removing the hard quasiparticle carrier.

Equivalently, the occupation can be embedded into two local variables:

$$\dot{n}_H = -\Gamma_H^{\text{occ}} Y,$$

$$\dot{Y} = \Lambda [(n_H - n_H^{\text{eq}}) - Y].$$

Eliminating Y gives an exponential retarded memory kernel.

At the benchmark,

$$\frac{\Gamma_H^{\text{occ}}}{\Lambda} = 3.48 \times 10^{-4}.$$

The maximum normalized difference between the memory evolution and the on-shell Markov limit is

$$3.45 \times 10^{-4}.$$

The equal-time spectral sum rule is recovered numerically:

$$\partial_t \rho_H(t, t'; \mathbf{p})|_{t=t'+} = 0.9999999998.$$

11. Cosmological cascade update

Using

$$\frac{\bar{\Gamma}_H^{\text{occ}}}{T} = 3.603 \times 10^{-4}$$

at

$$T_0 = 1.002 \times 10^8 \text{ GeV}$$

gives

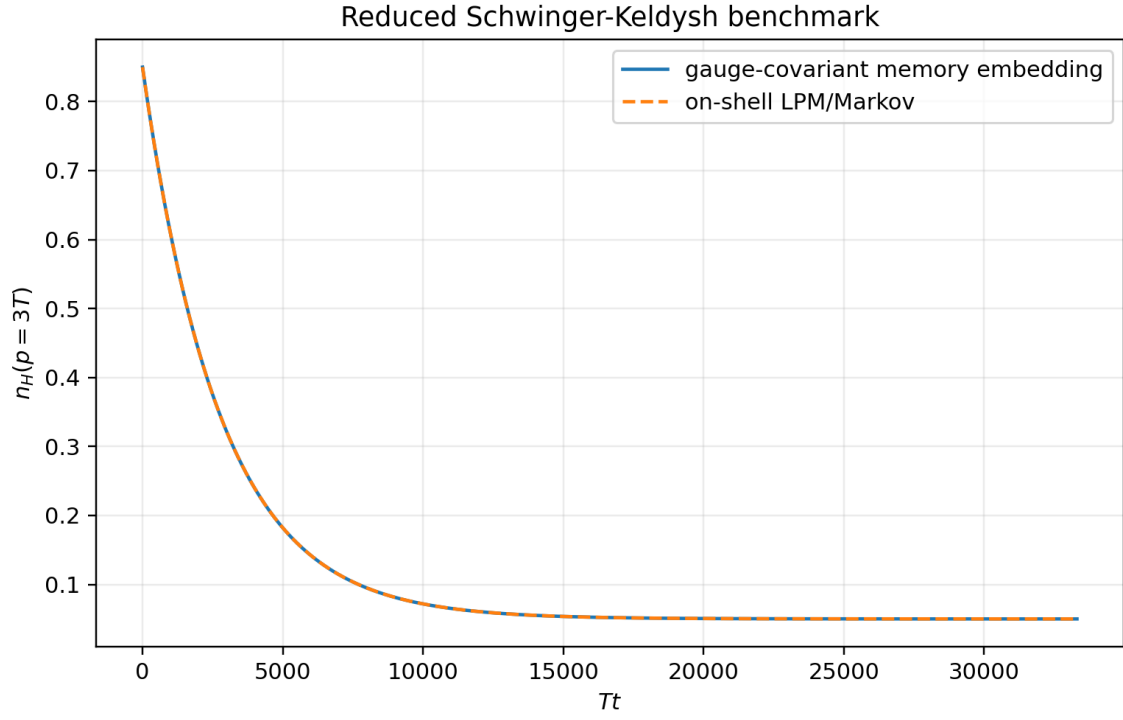


Figure 4: Finite-memory Schwinger-Keldysh reduction.

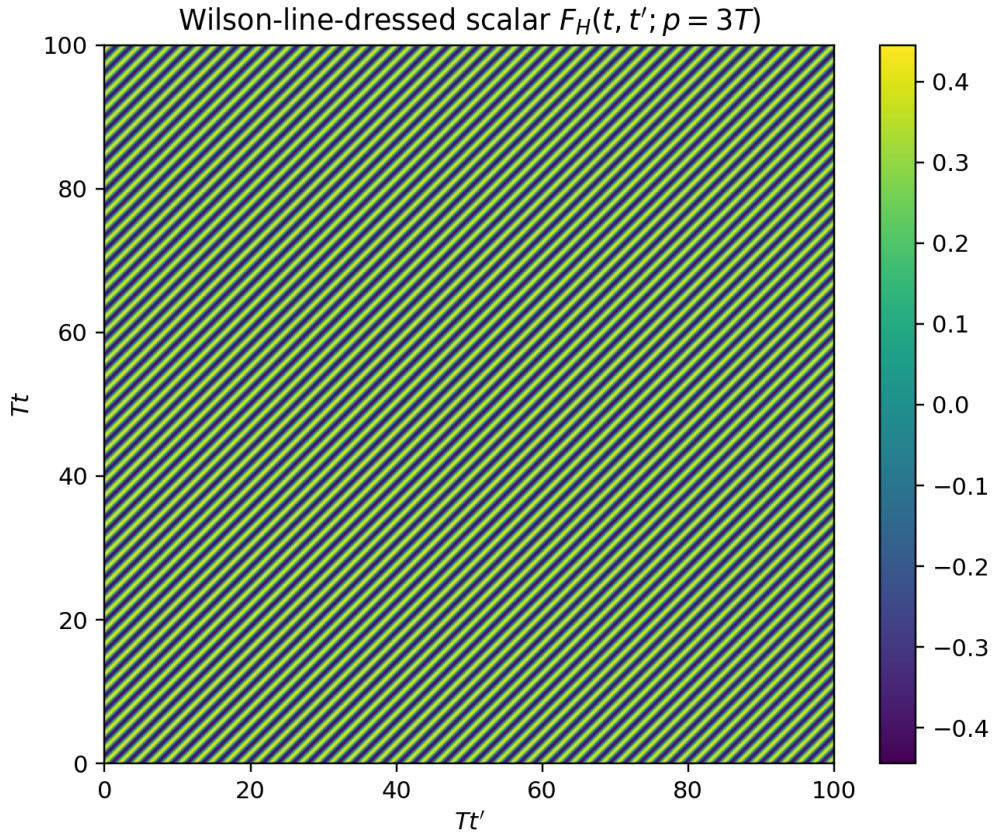


Figure 5: Wilson-line-dressed two-time scalar correlator.

$$\bar{\Gamma}_H^{\text{occ}} = 3.610 \times 10^4 \text{ GeV}.$$

For

$$\Gamma_R = 1.47850 \times 10^{-2} \text{ GeV},$$

$$\frac{\bar{\Gamma}_H^{\text{occ}}}{\Gamma_R} = 2.44 \times 10^6.$$

The maximum adiabatic lag is

$$\delta_{\text{ad}} \sim \frac{\Gamma_R}{\bar{\Gamma}_H^{\text{occ}}} = 4.09 \times 10^{-7}.$$

Thus

$$|\Delta B_5| < B_5 \delta_{\text{ad}} = 2.17 \times 10^{-9}.$$

The corrected values remain

$$B_5 = 0.00529888708$$

and

$$\frac{T_5}{T_0} = \frac{1}{4}$$

to the precision relevant for the cosmological vacuum-selection calculation.

Because the omitted hard $2 \leftrightarrow 2$ portal cuts increase the total rate, these lag and branching estimates are conservative.

12. Parameter dependence

A six-point direct scan at fixed

$$\alpha_s = 0.0393544$$

gives:

y_D	M_D/T	I_{LPM}	$\bar{\Gamma}_H^{\text{occ}}/T$
0.15	0.01	9.1540×10^{-4}	9.2684×10^{-5}
0.15	0.20	8.7887×10^{-4}	8.8986×10^{-5}
0.30	0.01	9.0115×10^{-4}	3.6497×10^{-4}
0.30	0.20	8.6566×10^{-4}	3.5059×10^{-4}
0.50	0.01	8.7106×10^{-4}	9.7994×10^{-4}
0.50	0.20	8.3766×10^{-4}	9.4237×10^{-4}

The integrated kernel changes only moderately with y_D and M_D/T over this range. The leading rate scaling remains approximately

$$\bar{\Gamma}_H^{\text{occ}} \propto y_D^2 T,$$

with calculable corrections through the thermal masses and formation dynamics.

13. Acceptance matrix

Target	Verdict	Result
Complete chiral/helicity source	PASS	Vector source $2\mathbf{P}$; two Higgs components and $N_c = 3$ included
Simultaneous $SU(3)$, $SU(2)$, $U(1)$ kernels	PASS	Exact Casimir and hypercharge interference coefficients
Thermal masses	PASS	Higgs thermal and Q, D asymptotic masses included
Thermal widths	PASS IN LPM SENSE	Self-energy and exchange pieces resummed jointly; gauge-invariant scalar width extracted
Published normalization test	PASS	0.898% difference from the published fit
Scalar retarded-self-energy matching	PASS INTEGRATED	Exact susceptibility-weighted on-shell identity
v1.4 portal normalization band	CLOSED	Direct LPM coefficient replaces factor-of-two band
Reduced gauge-covariant SK model	PASS AS BENCHMARK	Wilson-line dressing, retarded memory, and KMS noise structure
Complete leading-order portal rate	PARTIAL	Hard Yukawa-assisted $2 \leftrightarrow 2$ cuts remain
Full non-Abelian $3 + 1D$ 2PI/KB	OPEN	Ward-consistent two-time gauge-field evolution remains an HPC problem

14. What the result changes

The previous project logic was:

$$\text{QCD fast} \quad \text{and} \quad H \leftrightarrow qD \text{ uncertain by a factor of two.}$$

The new result is:

$\text{QCD fast} \quad \text{and} \quad H \leftrightarrow qD \text{ LPM normalization fixed to sub-percent validation accuracy.}$

The simultaneous gauge calculation also exposes a useful hierarchy:

$$SU(3)_c \gg U(1)_Y,$$

but

$$SU(2)_L$$

is not optional. It shifts the QCD-only portal width by roughly twenty percent and accounts for almost the entire difference between the QCD-only and full result.

The finite-memory correction is below 4×10^{-4} of the portal relaxation itself, while the portal relaxation is over two million times faster than the cosmological source. The reduced Schwinger-Keldysh model is therefore sufficiently accurate for the homogeneous cascade.

15. Remaining frontier

Before paying for a full non-Abelian $3 + 1D$ two-time simulation, two calculations now have higher scientific leverage.

15.1 Complete hard $2 \leftrightarrow 2$ portal self-energy

The complete leading-order qD contribution must add hard processes such as

$$G + H \leftrightarrow Q + D,$$

$$G + Q \leftrightarrow H + D,$$

$$Q + H \leftrightarrow G + D,$$

for

$$G = g, W, B,$$

including soft-fermion matching where the hard calculation becomes infrared sensitive.

This contribution is expected to increase the portal rate, not threaten the thermalisation conclusion. Its value matters for a genuinely complete pointwise scalar spectral function.

15.2 Momentum-resolved retarded kernel

The next correlator-level target is

$$\Pi_{H,qD}^R(k^0, |\mathbf{k}|)$$

on a two-dimensional frequency-momentum grid, with:

- the LPM collinear cut;
- hard $2 \leftrightarrow 2$ cuts;
- the KMS-related noise kernel;
- covariant Wigner transport;
- Ward-identity monitoring.

That table can then seed a reduced non-Abelian 2PI/Kadanoff-Baym implementation. Only after the reduced model reproduces the AMY/Kubo limit and the complete scalar spectral sum rules is a full $3 + 1D$ calculation worth its considerable computational bloodletting.

16. Strongest defensible conclusion

The complete collinear electroweak/Yukawa LPM kernel for $H \leftrightarrow Q_L D_R$ is finite, gauge organized, and numerically controlled.

Its exact integrated retarded-self-energy match gives $\bar{\Gamma}_H^{\text{occ}}/T = 3.603 \times 10^{-4}$.

This removes the old factor-of-two uncertainty and leaves the reheating hierarchy above 2.4×10^6 .

The portal does not threaten the cosmology. The unresolved problem is now narrower and more orthodox: calculate the hard $2 \leftrightarrow 2$ part of the same scalar self-energy and then lift the resulting complete kernel into a Ward-consistent two-time gauge theory.

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